

by using few resources. It's not just for more profit, but for more people, for masses.

Where does India stand with respect to the rest of the world in creating innovations?

It depends on what type of metrics you are using. We do not have a Samsung, LG or Nokia innovating in India, yet we have the largest number of mobile phone users in India. We are able to offer the cheapest plan and cheapest handset here, and yet make this business profitable. We do not have these technological innovations, but we have a business process innovation.

Similarly, take the example of Arvind Eye Care hospital. They do cataract eye surgery for only \$30 against \$3000 in the US. The doctors at this hospital perform 300,000 surgeries a year while maintaining the quality. The founder of the hospital was inspired by the McDonald's service efficiency and ensured the same at his hospital through the assembly line approach. This is work-flow innovation.

You have to enhance your metrics in order to understand innovation in the Indian context. If you evaluate by patents per capita or start-ups per capita, the population is so large that anything divided by a billion will give you a 'zero,' which is a wrong indicator. You have to relook at the innovations with a new concept and new model.

What is your take on technological innovations in India?

Today, if you look at any application—aerospace, defence, nanotechnology, biotechnology or automotive industry—you see developments all around.

We developed technologies under the condition of denial. We made our own supercomputer, aircraft and nuclear reactor when we did not have access to all these technologies from other countries.

I would like to quote the example of Saras—a 14-seater light transport aircraft project. It required around 15,000 components, of which only one component, that is the starter generator, was on the dual-use list. Due to some policies, this single component was denied to us by the US, leading to 18-month delay in the project. When Cray Supercomputer was denied to us, we developed our own Param Supercomputer. When Cryogenic engine was denied to Indian Space Research Organisation, they developed their own. So denial-driven innovation is not new to us.

What are the top five challenges to innovating in India?

The first challenge is resources. We don't have as much

money as many other countries. The budget of a company like Pfizer is around \$10 billion, whereas India's entire R&D budget is less than \$10 billion. So India does its research with a fraction of the money which is available to just one firm. This is a challenging situation in the competing world.

Second, we do not have critical mass. Suppose if I want to talk about some specific subject like Carbon nanotubes or graphene or Stem cell research, I do not find more than 5-10 people whom we can call world-class.

Lack of partnership between the industry and universities is another critical challenge. Many a time, inventions take place in universities but it requires a partnership with some company to convert an innovation into its successful use in practice. The industry needs to give students the research problems of their interest. Also, universities should choose problems that will have utility for the industry.

The fourth challenge is lack of awareness about patenting. If the innovation is not credited to your name, you will never be able to make profit out of it. You must own the rights of the innovations so that your work is respected, not copied, and credited to you.

The last challenge is the missing connect between Saraswati and Laxmi. We do Saraswati Puja and Laxmi

Puja separately. Globally, there are a number of scientists, even Nobel laureates, who are academicians but have their own companies. However, none of our academicians owns a company; people like Ashok Jhunjhunwala being an exception.

You should be able to create wealth out of the knowledge. When the Indian government funds a research, it's not your money, but the taxpayers' money. So when a new idea comes from the taxpayers' money, you should create wealth within the country for people of India.

What has been the government's take on innovations?

The government is very serious about this subject. Our prime minister has announced this decade (2011-20) as the decade of innovations. PM's National Innovation Council (www.innovationcouncil.gov.in/aboutus/aboutnuc.php) has been set up under the chairmanship of Sam Pitroda.

But we should not depend too much on the government. It's also the responsibility of the industry. We must cultivate a spirit where young people don't just seek jobs but carry an ambition to provide jobs by creating companies and enterprises. ●

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